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10 CFR 50.73

December 20, 2016  
NRC-16-0073

U. S. Nuclear Regulatory Commission  
Attention: Document Control Desk  
Washington, D.C. 20555-0001

Reference: Fermi 2  
NRC Docket No. 50-341  
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2016-012

Pursuant to 10 CFR 50.73(a)(2)(ii)(B), DTE Electric Company (DTE) is submitting LER No. 2016-012, Unanalyzed Condition for Control Rod Drop Accident at Low Reactor Power.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager –Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson  
Site Vice President

Enclosure: Licensee Event Report No. 2016-012, Unanalyzed Condition for Control Rod Drop Accident at Low Reactor Power

cc: NRC Project Manager  
NRC Resident Office  
Reactor Projects Chief, Branch 5, Region III  
Regional Administrator, Region III  
Michigan Public Service Commission  
Regulated Energy Division (kindschl@michigan.gov)

**Enclosure to  
NRC-16-0073**

**Fermi 2 NRC Docket No. 50-341  
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2016-012, Unanalyzed Condition for Control  
Rod Drop Accident at Low Reactor Power**



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

## 1. FACILITY NAME

Fermi 2

## 2. DOCKET NUMBER

05000 341

## 3. PAGE

1 OF 5

## 4. TITLE

Unanalyzed Condition for Control Rod Drop Accident at Low Reactor Power

| 5. EVENT DATE     |     |      | 6. LER NUMBER                                                                                 |                                             |          | 7. REPORT DATE                                         |     |                                               | 8. OTHER FACILITIES INVOLVED |               |
|-------------------|-----|------|-----------------------------------------------------------------------------------------------|---------------------------------------------|----------|--------------------------------------------------------|-----|-----------------------------------------------|------------------------------|---------------|
| MONTH             | DAY | YEAR | YEAR                                                                                          | SEQUENTIAL NUMBER                           | REV. NO. | MONTH                                                  | DAY | YEAR                                          | FACILITY NAME                | DOCKET NUMBER |
| 11                | 02  | 2016 | 2016                                                                                          | 012                                         | 00       | 12                                                     | 20  | 2016                                          | N/A                          | 05000         |
| 9. OPERATING MODE |     |      | 11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply) |                                             |          |                                                        |     |                                               |                              |               |
| 1                 |     |      | <input type="checkbox"/> 20.2201(b)                                                           | <input type="checkbox"/> 20.2203(a)(3)(i)   |          | <input type="checkbox"/> 50.73(a)(2)(ii)(A)            |     | <input type="checkbox"/> 50.73(a)(2)(viii)(A) |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2201(d)                                                           | <input type="checkbox"/> 20.2203(a)(3)(ii)  |          | <input checked="" type="checkbox"/> 50.73(a)(2)(ii)(B) |     | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(1)                                                        | <input type="checkbox"/> 20.2203(a)(4)      |          | <input type="checkbox"/> 50.73(a)(2)(iii)              |     | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(2)(i)                                                     | <input type="checkbox"/> 50.36(c)(1)(i)(A)  |          | <input type="checkbox"/> 50.73(a)(2)(iv)(A)            |     | <input type="checkbox"/> 50.73(a)(2)(x)       |                              |               |
| 10. POWER LEVEL   |     |      | <input type="checkbox"/> 20.2203(a)(2)(ii)                                                    | <input type="checkbox"/> 50.36(c)(1)(ii)(A) |          | <input type="checkbox"/> 50.73(a)(2)(v)(A)             |     | <input type="checkbox"/> 73.71(a)(4)          |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(2)(iii)                                                   | <input type="checkbox"/> 50.36(c)(2)        |          | <input type="checkbox"/> 50.73(a)(2)(v)(B)             |     | <input type="checkbox"/> 73.71(a)(5)          |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(2)(iv)                                                    | <input type="checkbox"/> 50.46(a)(3)(ii)    |          | <input type="checkbox"/> 50.73(a)(2)(v)(C)             |     | <input type="checkbox"/> 73.77(a)(1)          |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(2)(v)                                                     | <input type="checkbox"/> 50.73(a)(2)(i)(A)  |          | <input type="checkbox"/> 50.73(a)(2)(v)(D)             |     | <input type="checkbox"/> 73.77(a)(2)(i)       |                              |               |
|                   |     |      | <input type="checkbox"/> 20.2203(a)(2)(vi)                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)  |          | <input type="checkbox"/> 50.73(a)(2)(vii)              |     | <input type="checkbox"/> 73.77(a)(2)(ii)      |                              |               |
|                   |     |      | <input type="checkbox"/> 50.73(a)(2)(i)(C)                                                    | <input type="checkbox"/> OTHER              |          | Specify in Abstract below or in NRC Form 366A          |     |                                               |                              |               |

## 12. LICENSEE CONTACT FOR THIS LER

## LICENSEE CONTACT

Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing

## TELEPHONE NUMBER (Include Area Code)

(734) 586-5076

## 13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

| CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX | CAUSE | SYSTEM | COMPONENT | MANU-FACTURER | REPORTABLE TO EPIX |
|-------|--------|-----------|---------------|--------------------|-------|--------|-----------|---------------|--------------------|
| N/A   | N/A    | N/A       | N/A           | N/A                | N/A   | N/A    | N/A       | N/A           | N/A                |

## 14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO

## 15. EXPECTED SUBMISSION DATE

| MONTH | DAY | YEAR |
|-------|-----|------|
|       |     |      |

## ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On November 2, 2016, a non-conservatism in the current Fermi 2 design and licensing basis of the control rod drop accident (CRDA) was identified. The current analysis assumes that the fission products released as a result of a CRDA are transported to the condenser and then released from there. An unanalyzed condition was identified where a forced release from the Gland Seal Exhausters (GSEs) could occur which would exceed dose limits in 10 CFR 100.11 and Standard Review Plan 6.4 when operating at low reactor power conditions. The postulated fuel damage and associated release is only credible for low power conditions of less than 10% power. Fermi 2 was at 97% power at the time of discovery and, therefore, the condition did not exist. However, a review of past operating conditions identified five instances in the past three years where Fermi 2 operated at less than 10% power. Based on a subsequent engineering analysis, for each of those five instances, plant conditions and available equipment were such that dose limits would have been met even if a postulated CRDA had occurred during those times; therefore, there was no adverse impact to public health and safety or to plant employees. There were no radiological releases. The cause of the event was that the original design and licensing basis analysis of the CRDA did not evaluate the direct release of steam from the GSEs, and, therefore, the plant was not designed to mitigate the potential consequences associated with that release path. For corrective actions, the plant startup checklist was revised to impose restrictions on future startups based on the engineering analysis. Fermi 2 plans to pursue modifications which would eliminate this potential release path.



**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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| 1. FACILITY NAME |  | 2. DOCKET NUMBER |     | 3. LER NUMBER |                   |         |
|------------------|--|------------------|-----|---------------|-------------------|---------|
| Fermi 2          |  | 05000-           | 341 | YEAR          | SEQUENTIAL NUMBER | REV NO. |
|                  |  |                  |     | 2016          | 012               | 00      |

**NARRATIVE****INITIAL PLANT CONDITIONS**

Mode – 1

Reactor Power – 97 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

**DESCRIPTION OF THE EVENT**

On November 2, 2016, while performing a re-evaluation of the radiological consequences of the control rod drop accident (CRDA) in support of a potential license amendment, DTE Electric Company (DTE) identified a non-conservatism in the current Fermi 2 design and licensing basis of the CRDA. The non-conservatism was identified at approximately 1030 EDT while operating in MODE 1 at 97% power.

The current CRDA analysis, which is based on Standard Review Plan (SRP) 15.4.9, postulates that radioactive fission products are released as a result of the control rod drop and the released activity transports directly to the condenser and is released at ground level from the turbine building at the rate of 1% of the condenser volume per day for one day. Doses reported in the current licensing basis analysis demonstrate offsite radiological consequences are well within (less than 25% of) the Title 10 of the Code of Federal Regulations (CFR) Part 100.11 siting criteria. The current licensing basis was approved without specific evaluation of Main Control Room (MCR) [[NA]] personnel dose.

Per Updated Final Safety Analysis Report (UFSAR) Section 6.4, the functional design of the habitability systems ensure compliance with 10 CFR 50 Appendix A General Design Criterion (GDC) 19 and that "[u]nder any design-basis conditions, the environment within the main control room is safe, with personnel protected from radiation, fire, toxic gases, and noxious substances." Sufficient radiation shielding and air filtration are provided to ensure that radiation exposures of MCR personnel do not exceed 5 rem whole-body, or its equivalent to any part of the body, for the duration of a design basis accident (DBA). For the DBA Loss of Coolant Accident (LOCA) and the Fuel Handling Accident, the dose to main control room personnel does not exceed 5 rem total effective dose equivalent (TEDE).

The Control Room Emergency Filtration (CREF) [[VI]] system is an Engineered Safety Feature (ESF) designed to limit radiation exposure to personnel occupying the MCR to less than 5 rem over the duration of a DBA as required by 10 CFR 50.67 and 10 CFR 50 Appendix A Criterion 19. This function is accomplished by 1) isolating normal unfiltered intakes, 2) directing emergency makeup air through charcoal and High Efficiency Particulate Air (HEPA) filters [[FLT]], 3) re-circulating control room and makeup air through HEPA and charcoal filters, and 4) maintaining the Control Room Envelope (CRE) under a positive pressure with respect to the outside environs to prevent the in-leakage of unfiltered outside air. CRE integrity is necessary to minimize potential unfiltered in-leakage thereby supporting the operation of the Control Center Heating Ventilating and Air Conditioning (CCHVAC) [[VI]] and CREF systems in their function to protect the health and safety of control room operators post-accident.

As described above, the radiological consequences of the CRDA were re-evaluated to support a potential license amendment. The results of the new analysis, which were obtained in accordance with current regulatory guidance that has been updated from the original NUREG 75/087 Section 15.4.9 Fermi 2 licensing basis, predict onsite and offsite radiological consequences that exceed the current applicable regulatory dose limits in 10 CFR 100.11 and SRP 6.4 (GDC 19). The increase in calculated radiological consequences is attributable to the consideration of a forced release via the



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|                  |                  | 2016          | 012               | 00      |

**NARRATIVE**

Gland Sealing Steam Exhausters (GSEs). This release path was not explicitly evaluated under the original plant design and licensing basis; however, its consideration is consistent with current NRC Regulatory Guides (RGs) 1.183 (regarding Alternate Source Term (AST) methods) and 1.195 (regarding non-AST methods) as well as industry precedent. Under normal full power operating conditions, the fraction of total steam production delivered to the GSEs is small (less than 0.2%). However, at low reactor power, when the consequences of a postulated CRDA are credible, the fraction of gland sealing steam relative to total steam production is relatively large (2-20%). Furthermore, this path does not benefit from iodine holdup and NRC precedent does not allow credit for iodine washout for this path. Consequently, the modeled release is greater than regulatory dose limits. Performed as directed by SRP 15.4.9, the current plant CRDA analysis evaluates only the consequences associated with a 1% volume/day release from the main condenser and includes credit for iodine retention in the vessel and main condenser. These are removal mechanisms that significantly reduce the consequences of the release. The only forced release path that is considered is that associated with the mechanical vacuum pumps (MVPs). However, the MVPs are provided with an automatic trip on Offgas 2-Minute Delay Pipe HI-HI Radiation that is credited to mitigate consequences from this forced release path. The release path from the GSEs was not evaluated and an automatic trip of the GSEs was not provided. Although UFSAR Section 15.4.9 does not evaluate or report MCR operator doses, UFSAR Section 6.4 states that the habitability design of the main control room is adequate to protect the control room personnel for all accidents.

There is no degraded component. Rather, the condition is being treated as the result of an inadequate design and licensing basis. The results of the updated analysis establish that in the absence of an installed means of rapidly and automatically terminating the release associated with the GSEs, regulatory dose limits are exceeded unless credit is taken for assumptions and features not currently part of the licensing basis, such as more realistic input assumptions regarding core source term and existing installed methods of mitigation, such as Main Steam Isolation Valve (MSIV) [[ISV]] isolation and operation of the CREF system.

The condition is specific to the analysis of the CRDA for which the postulated fuel damage occurs at time zero and resultant activity instantaneously transports to the coolant in the reactor vessel. Reactor fuel damage associated with other accidents such as the DBA LOCA occurs on a time scale that is longer than the time required for isolation of the steam lines and a release via the GSEs is not postulated to occur. On this basis, the condition is only applicable to the CRDA. In addition, the postulated fuel damage resulting from the CRDA is only credible during low-power (less than 10%) MODE 1 and 2 operation during a plant startup. With increasing power, the severity of the rod drop reactivity insertion is reduced such that the expected peak enthalpy is much less than the design limit at 10% power even if the worst conceivable or maximum worth control rod is dropped from the core. Consistent with this expected power dependence of the consequences of the CRDA, the Rod Worth Minimizer is required to be operable for a reactor power less than or equal to 10% (per Technical Specification Table 3.3.2.1-1). The lower the assumed power level, the larger the source term fraction associated with gland sealing steam. The results of the updated CRDA analysis demonstrate that the consequences of CRDA lessen as the assumed power level increases due to the fact that as steam production increases, an increasing fraction of the source term is directed to the condenser where it is held up and decayed, and released at a reduced rate.

At the time of the discovery, Fermi 2 was operating at 97% power. Therefore, the plant condition was bounded by the design and licensing basis such that the condition did not exist at the time of discovery. No immediate actions were required. However, Fermi 2 had operated at low reactor power levels during startup several times in the past three years. A past operability review for the three years prior to November 2, 2016 was performed to identify when Fermi 2 had operated at the relevant power levels (i.e. less than 10%). The instances are listed below:



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| Fermi 2          |  | 05000-           | 341 | YEAR<br>2016  | SEQUENTIAL<br>NUMBER<br>012 | REV<br>NO.<br>00 |

**NARRATIVE**

- 1) At approximately 0130 EDT on March 28, 2014, Fermi 2 entered MODE 2 following a refueling outage. Power remained below 10% during startup until after approximately 0730 EDT on March 31, 2014. The duration in the relevant power range was approximately three days.
- 2) At approximately 1730 EDT on April 21, 2014, Fermi 2 entered MODE 2 following an outage. Power remained below 10% during startup until after approximately 0730 EDT on April 23, 2014. The duration in the relevant power range was approximately two days.
- 3) At approximately 0730 EDT on April 3, 2015, Fermi 2 entered MODE 2 following an outage. Power remained below 10% during startup until after approximately 0730 EDT on April 4, 2015. The duration in the relevant power range was approximately one day.
- 4) At approximately 1300 EST on November 25, 2015, Fermi 2 entered MODE 2 following a refueling outage. Power remained below 10% during startup until after approximately 0730 EST on November 28, 2015. The duration in the relevant power range was approximately three days.
- 5) At approximately 1630 EDT on May 12, 2016, Fermi 2 entered MODE 2 following an outage. Power remained below 10% during startup until after approximately 1930 EDT on May 13, 2016. The duration in the relevant power range was approximately one day.

The five periods identified above where Fermi 2 was operating in MODE 1 or 2 below 10% power represent potential unanalyzed conditions where the occurrence of a CRDA during those periods could have resulted in offsite and main control room doses exceeding regulatory limits. Therefore, this Licensee Event Report (LER) is being made under 10 CFR 50.73(a)(2)(ii)(B) as an "event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety." An 8-hour non-emergency Event Notification (EN 52342) was previously made under the corresponding requirement in 10 CFR 50.72(b)(3)(ii)(B).

Subsequent to discovery but prior to the date of this LER, Fermi 2 entered MODE 2 on November 11, 2016 following an outage that began on November 7, 2016. This startup was performed in accordance with plant procedures which had been revised based on an engineering analysis, described in detail below, and was, therefore, not an unanalyzed condition.

**SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS**

An engineering analysis was performed to evaluate the safety consequences and implications. The analysis took credit for assumptions and features that, while justified, are not currently part of the licensing basis. Regarding offsite doses, the analysis determined that crediting automatic closure of the MSIVs during the CRDA is sufficient to ensure offsite doses remain within the limits of 10 CFR 100.11 and the more restrictive SRP 15.4.9 requirement that consequences for the CRDA remain less 25% of the Part 100 limits. Crediting MSIV closure is reasonable because it is an available ESF controlled by Technical Specifications (TS 3.3.6.1). Additional margin to these limits exists when typical post-shutdown decay of the core source term and post-scrum reduction in steam flow are considered. Regarding MCR doses, the analysis took credit for automatic closure of the MSIVs as described previously and also for operation of the CREF since it is also an available ESF controlled by Technical Specifications (TS 3.7.3). Credit for these features alone was insufficient to ensure operator consequences within SRP 6.4. However, credit for these features in conjunction with either of the following conservatisms was sufficient to ensure that current NRC acceptance limits were met:

- Credit for the last as-found control room unfiltered in-leakage, which is nearly one-half the current 173 cfm design value
- Source term decay due to normal post-shutdown delay greater than 5.1 days in startup (normally days to weeks for any shutdown from full power)



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|                  |  |                  |  | 2016          | 012                  | 00         |

**NARRATIVE**

Note that the 30-day calculated MCR post-CRDA dose is subject to a current regulatory acceptance limit of 5 rem whole body/30 rem thyroid based on Revision 2 of SRP 6.4. Revision 3 of SRP 6.4 currently allows adoption of a higher thyroid dose limit of 50 rem for current operating reactors conforming to and implementing the guidance of RG 1.195 in conjunction with RG 1.196. Since the analysis was performed in accordance with those RGs, the higher 50 rem thyroid dose limit was adopted for the analysis.

This LER identified five instances in the past three years that represented unanalyzed conditions. In each of those instances, the plant was in the relevant power range (i.e. less than 10%) during startup following an outage. The duration that the plant was shut down prior to startup was greater than 5.1 days for each of the five instances. Therefore, the actual plant condition for these five instances met the second bullet identified above such that MCR doses for a postulated CRDA would have met NRC acceptance limits. Note that credit for either one of the bulleted items is sufficient to provide adequate mitigation; however, both may be in effect, which provides additional margin. In addition, washout of iodine in the gland seal exhaust condenser, which would also significantly reduce the release of iodine, was not credited in the evaluation.

In conclusion, the engineering analysis determined that even if a postulated CRDA had occurred during the three years prior to the discovery of this unanalyzed condition, all relevant dose acceptance criteria for both offsite and MCR doses would have been met based on the plant condition and available ESF equipment at the time. Therefore, there was no adverse impact to public health and safety or to plant employees. There were no radiological releases.

**CAUSE OF THE EVENT**

The cause was that the original design and licensing basis analysis of the Fermi 2 DBA CRDA described in UFSAR Section 15.4.9 was performed in accordance with the prescribed methods and assumptions of NRC SRP 15.4.9 that did not evaluate the direct release of nuclear steam from the GSEs as significant. As a result, the plant was not designed to mitigate the potential consequences associated with this release path.

**CORRECTIVE ACTIONS**

As discussed previously, no immediate action was required to restore compliance since the condition did not exist at the time of discovery. As a long-term corrective action, Fermi 2 plans to implement modifications to eliminate this potential CRDA release path. To prevent future occurrence of an unanalyzed condition in the interim until the modification is complete, the Post Outage Startup Checklist was revised on November 10, 2016, to impose restrictions on the plant startup based on the engineering analysis described above. This checklist was utilized during the plant startup that was begun on November 11, 2016.

**PREVIOUS OCCURRENCES**

No similar previous occurrences were identified.